01 — Bench input and logic supplies

12 V input / fuse / reverse polarity
5 V buck + 3.3 V LDO
Separate bridge power link JP1

File: 01-power.kicad_sch

02 — STM32G431CBT6, LQFP48

STM32G431CBT6 / LQFP48
TIM1 PWM + ADC1/ADC2 feedback
HSI clock / SWD / reset / BOOT0

File: 02-mcu.kicad_sch

03 — DRV8323RS gate driver and current feedback

DRV8323RS / VQFN48 + EP
SPI / charge pump / three CSAs
ENABLE and nFAULT / BKIN

File: 03-driver.kicad_sch

04 — Three half bridges and Kelvin shunts

Six LFPAK56 MOSFETs
Three 20 mOhm Kelvin shunts
Motor phases A / B / C

File: 04-inverter.kicad_sch

05 — Bench interfaces and measurement points

Encoder / SWD / UART / DShot
VBUS divider / temperature NTC
3.3 V signal interfaces

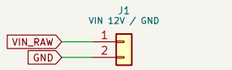
File: 05-interfaces.kicad_sch

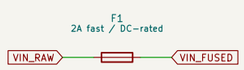
06 — Test points and mounting

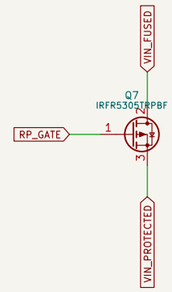
36 measurement pads
Four M3 mounting holes
Common GND, no isolation

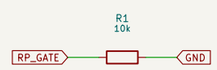
File: 06-testpoints.kicad_sch

12 V nominal, 9–16 V input; 1 A phase peak design target



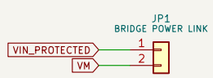






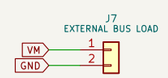


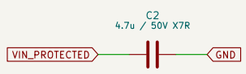


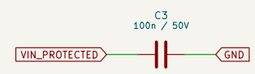


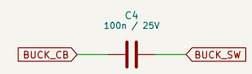




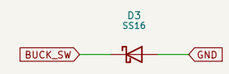


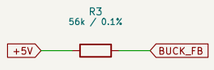








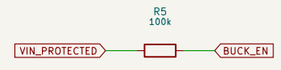


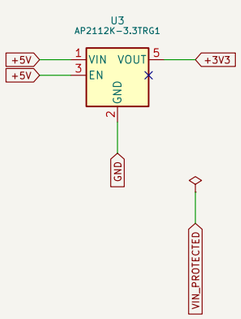


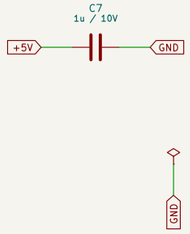


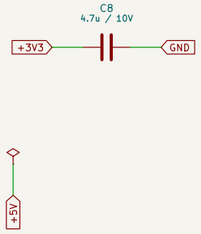


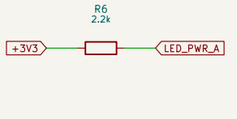






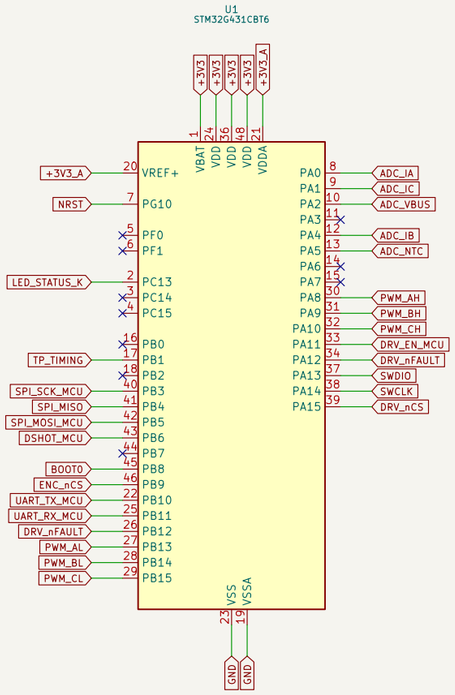








U1 pin assignment follows hardware/dev-board/cubemx/CubeMX.ioc

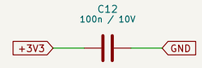




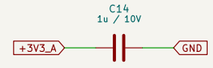










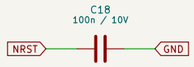










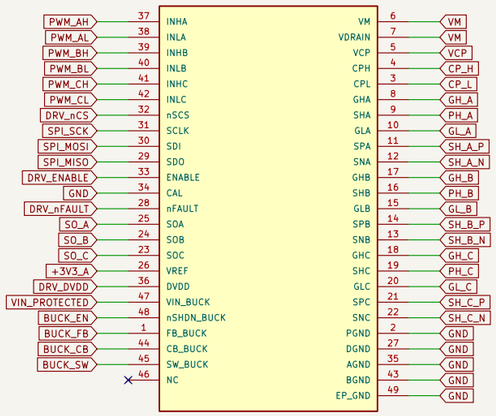








U2: 48 pins + exposed ground pad; charge pump for motor high-side gates

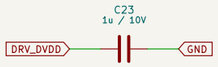
















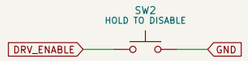






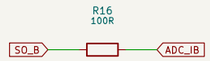












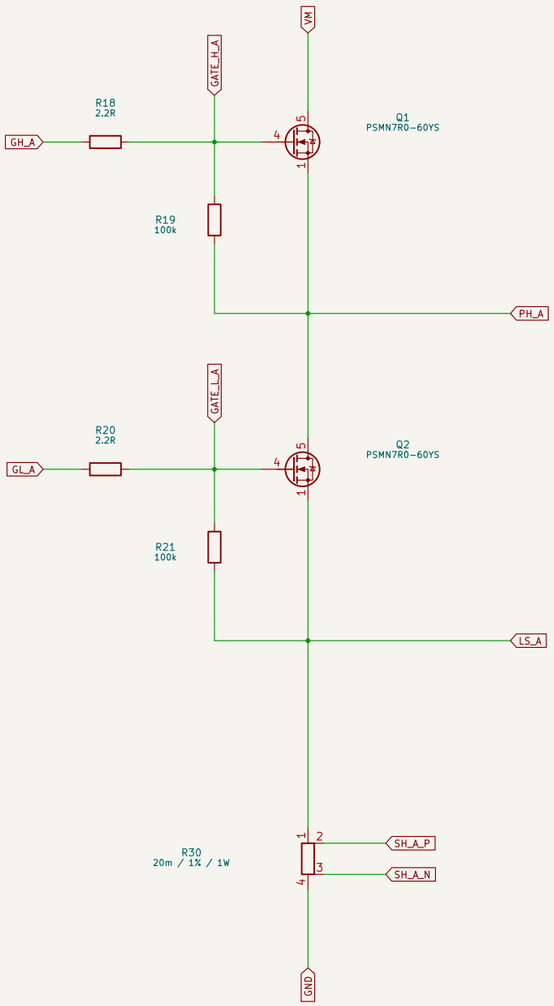




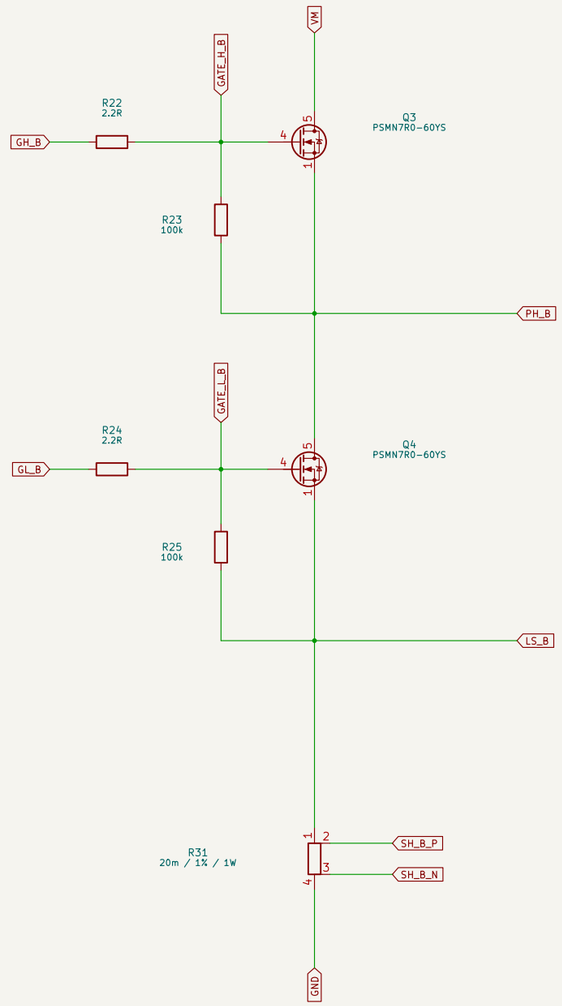


SINGLE MOTOR — 12 V bench / 1 A phase peak design target

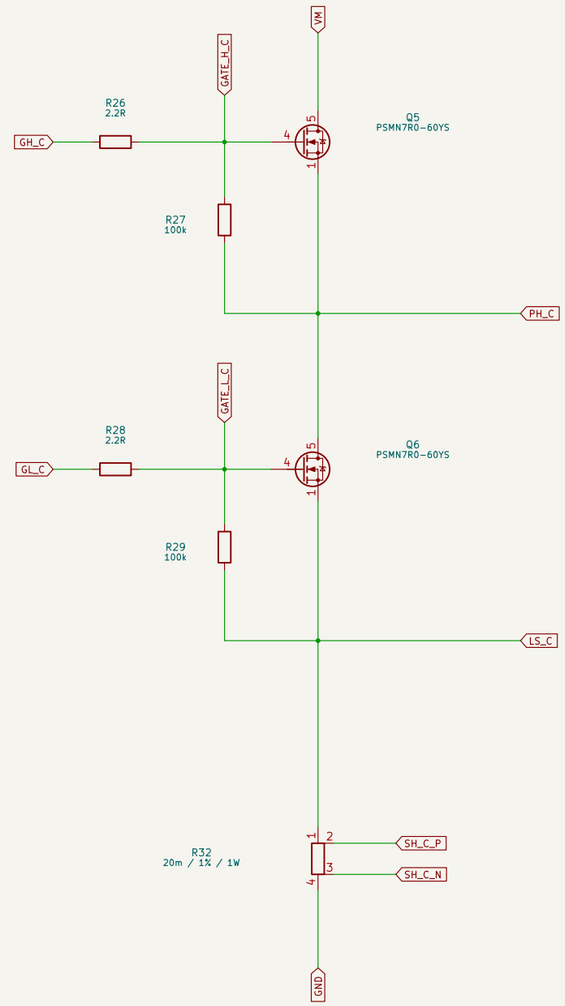
PHASE A

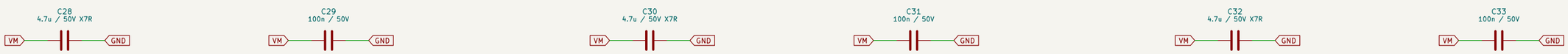


PHASE B

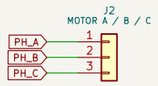


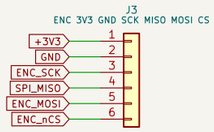
PHASE C

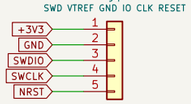


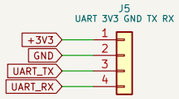


All signal headers use 3.3 V logic; target-powered voltage pins are OUTPUT / VTREF only













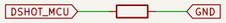




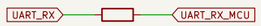






















Measurement access — probe ground clips connect ONLY to GND

TP1
GND
GND

TP2
GND
GND

TP3
GND
GND

TP4
VIN_PROTECTED
VIN_PROTECTED

TP5
VM
VM

TP6
+5V
+5V

TP7
+3V3
+3V3

TP8
+3V3_A
+3V3_A

TP9
DRV_ENABLE
DRV_ENABLE

TP10
DRV_nFAULT
DRV_nFAULT

TP11
TP_TIMING
TP_TIMING

TP12
SPI_MISO
SPI_MISO

TP13
PWM_AH
PWM_AH

TP14
PWM_AL
PWM_AL

TP15
PWM_BH
PWM_BH

TP16
PWM_BL
PWM_BL

TP17
PWM_CH
PWM_CH

TP18
PWM_CL
PWM_CL

TP19
ADC_IA
ADC_IA

TP20
ADC_IB
ADC_IB

TP21
ADC_IC
ADC_IC

TP22
ADC_VBUS
ADC_VBUS

TP23
ADC_NTC
ADC_NTC

TP24
DRV_DVDD
DRV_DVDD

TP25
GATE_H_A
GATE_H_A

TP26
PH_A
PH_A

TP27
GATE_L_A
GATE_L_A

TP28
LS_A
LS_A

TP29
GATE_H_B
GATE_H_B

TP30
PH_B
PH_B

TP31
GATE_L_B
GATE_L_B

TP32
LS_B
LS_B

TP33
GATE_H_C
GATE_H_C

TP34
PH_C
PH_C

TP35
GATE_L_C
GATE_L_C

TP36
LS_C
LS_C

H1
M3 NPTH

H2
M3 NPTH

H3
M3 NPTH

H4
M3 NPTH